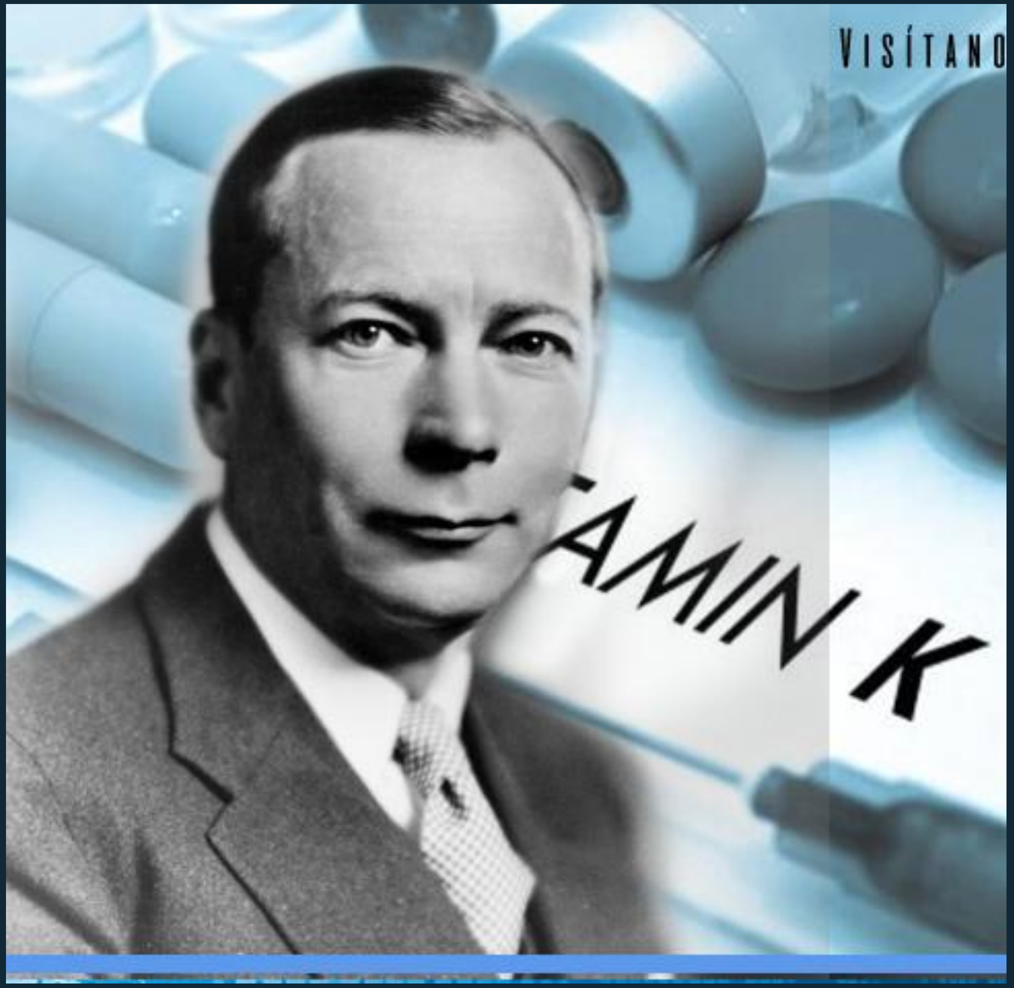
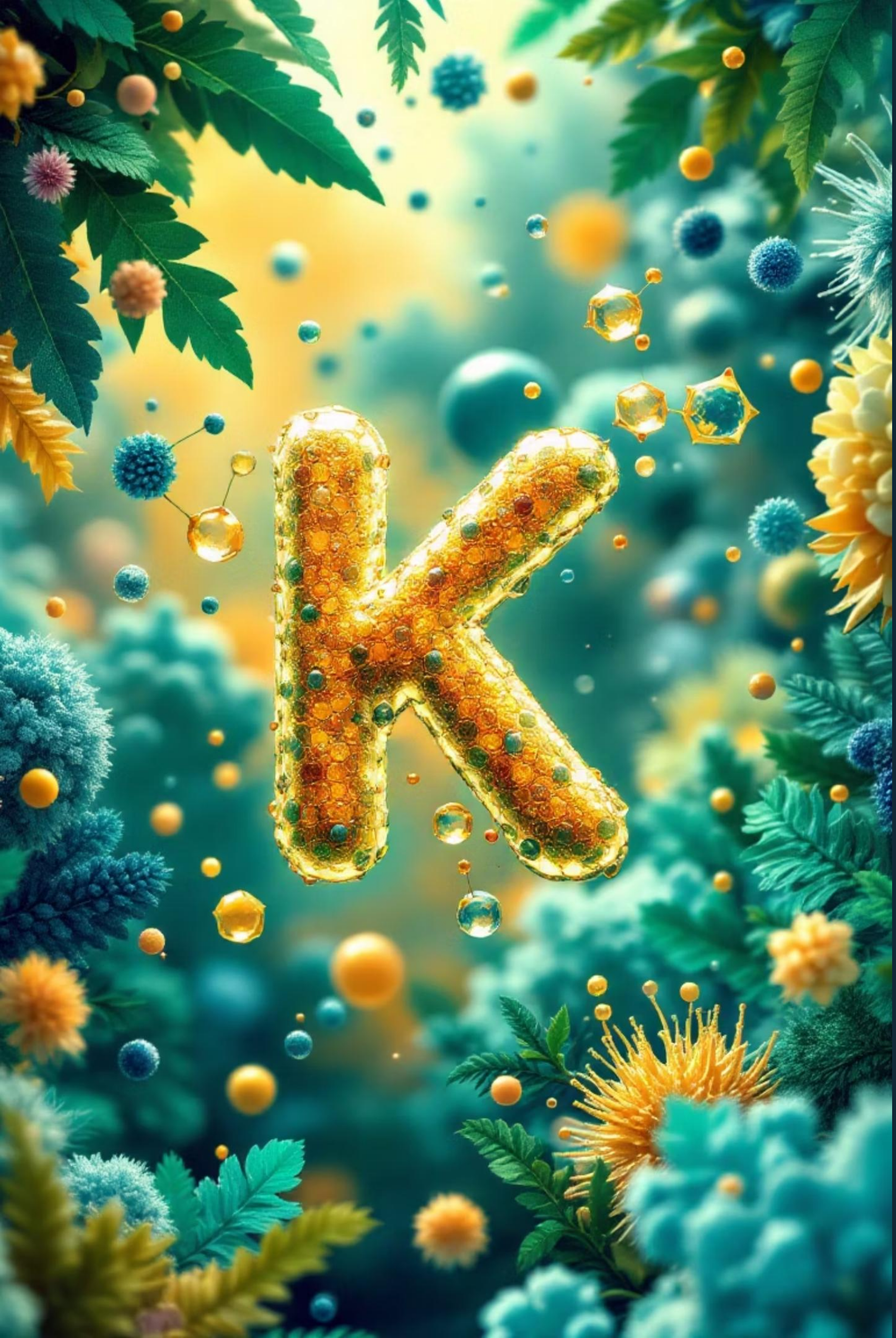
VITAMIN K

CONTENTS

INTRODUCTION

- Vitamin K is a fat-soluble vitamin essential for blood coagulation (from German "Koagulation").
- Discovered in 1929 by Dr. Henrik Dam; Nobel Prize awarded in 1943 to Dam and Edward Doisy for its discovery and structure.
- Unique as the only fat-soluble vitamin with a specific coenzyme function in post-translational modification of proteins.
- Vital for synthesizing clotting factors and bone metabolism.





TYPES OF VITAMIN K

VITAMIN K1 (PHYLLOQUINONE)

Found in green leafy plants; primary dietary source. Essential for photosynthesis in plants.

VITAMIN K2 (MENAQUINONE)

Produced by intestinal bacteria; also found in fermented foods and animal products; longer half-life and better absorption.

VITAMIN K3 (MENADIONE)

Synthetic form; water-soluble precursor converted in the body to active forms. Not used in human supplements due to toxicity.

SOURCES

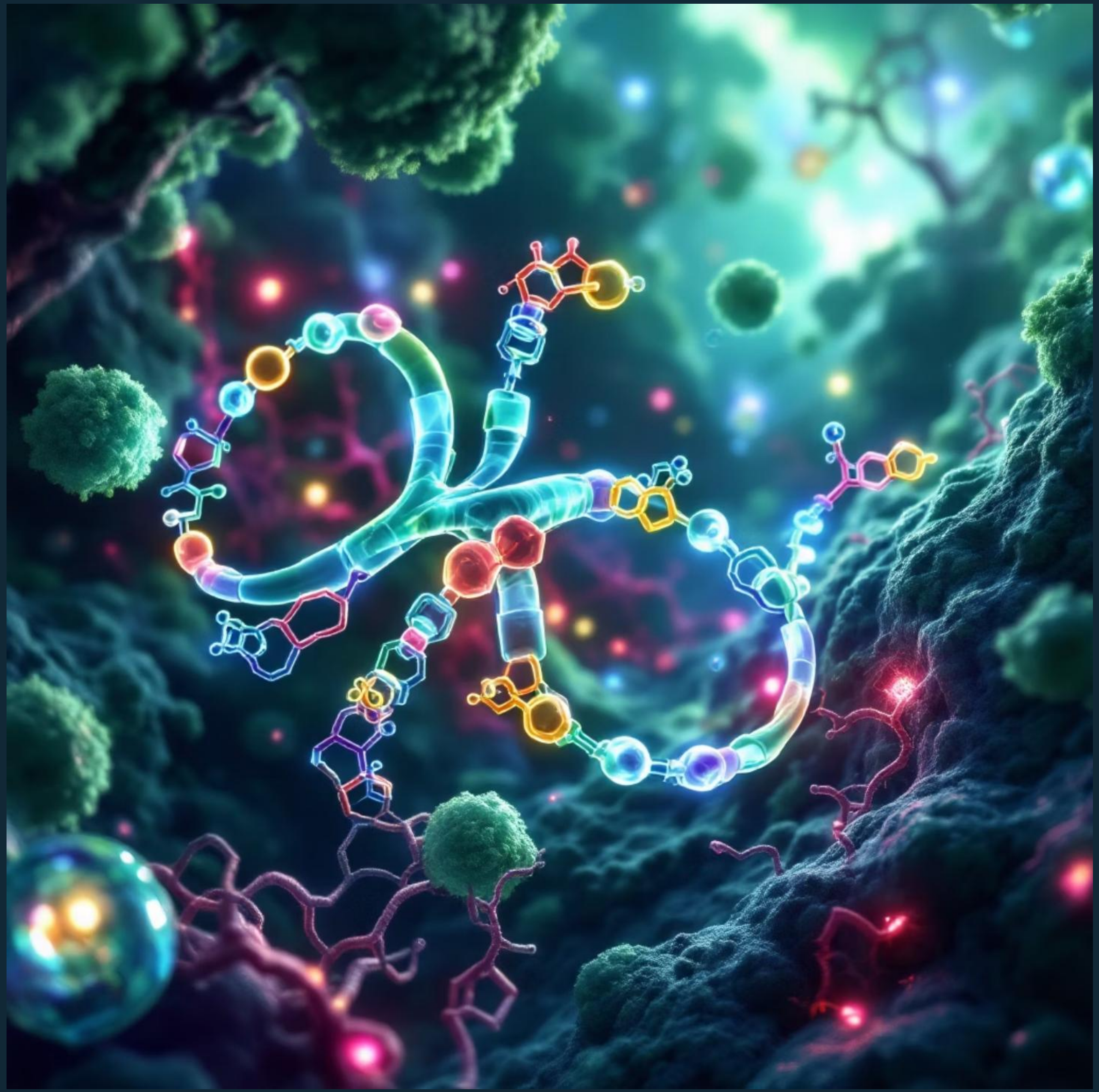
- Leafy greens (spinach, kale, broccoli)
- Vegetable oils
- Meat, eggs, fish
- Fermented foods (natto)
- Gut synthesis






FUNCTIONS

- Acts as a coenzyme for gamma-glutamyl carboxylase enzyme in the liver.
- Enables carboxylation of glutamic acid residues on clotting factors II (prothrombin), VII, IX, and X, activating them for blood coagulation, a process known as the Vitamin K cycle.
- Supports bone health by carboxylating osteocalcin, a calcium-binding protein important for bone mineralization and strength.
- Involved in vascular health by regulating proteins that prevent arterial calcification, thus reducing the risk of cardiovascular disease.



DEFICIENCY

- Rare in healthy adults due to adequate dietary intake and efficient bacterial synthesis in the gut.
- Causes include fat malabsorption disorders (e.g., cystic fibrosis, Crohn's disease), prolonged antibiotic use disrupting gut flora, severe liver disease, and newborns’ limited stores and underdeveloped gut flora.
- Symptoms: increased bleeding tendency, easy bruising, recurrent nosebleeds, bleeding gums, and in infants, a severe condition known as hemorrhagic disease of the newborn (VKDB).
- Deficiency leads to impaired synthesis of active clotting factors, resulting in significantly prolonged clotting time and potential for severe hemorrhage.

HEALTH BENEFITS

BLOOD CLOTTING

Prevents excessive bleeding by ensuring proper synthesis and activation of vital blood clotting factors.

BONE STRENGTH

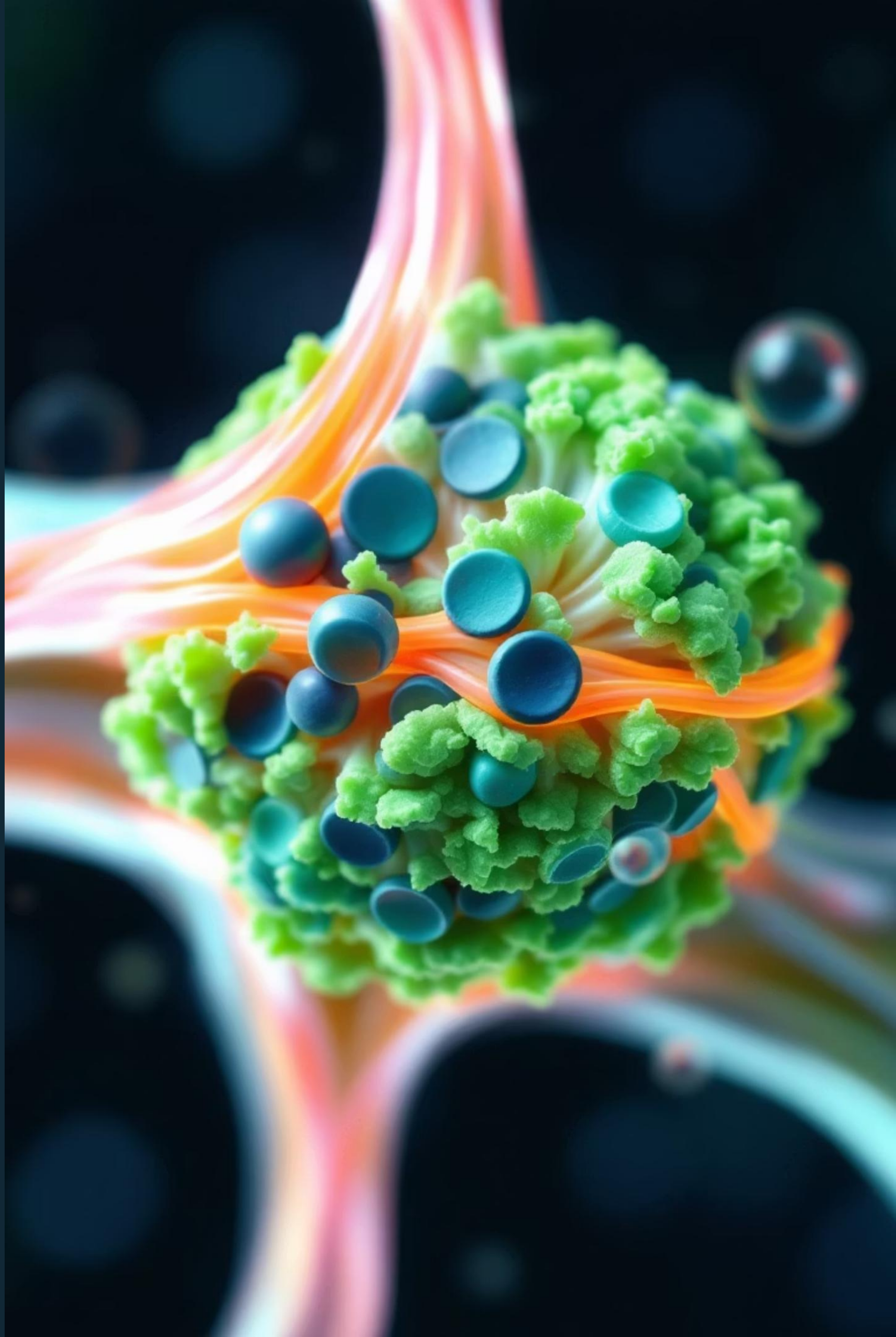
Supports bone mineralization and reduces the risk of osteoporosis by activating osteocalcin, a key protein for calcium binding in bones.

CARDIOVASCULAR HEALTH

May reduce arterial calcification and maintain arterial elasticity, thereby lowering the risk of cardiovascular disease.

CELLULAR REGULATION

Emerging research suggests potential roles in cell growth regulation and inhibitory effects on cancer progression.



RECOMMENDED INTAKE & SIDE EFFECTS

RECOMMENDED DAILY ALLOWANCE (RDA):

- Adults: 70–140 µg/day (varies by age, sex, and life stage). Typical recommendations are 90 µg for adult women and 120 µg for adult men.
- Infants and children require lower amounts; newborns often receive a prophylactic vitamin K injection at birth to prevent hemorrhagic disease.

AGE	MALE	FEMALE	PREGNANCY	LACTATION
Birth to 6 months	2.0 mcg	2.0 mcg	--	--
7-12 months	2.5 mcg	2.5 mcg	--	--
1-3 years	30 mcg	30 mcg	--	--
4-8 years	55 mcg	55 mcg	--	--
9-13 years	60 mcg	60 mcg	--	--
14-18 years	75 mcg	75 mcg	75 mcg	75 mcg
19+ years	120 mcg	90 mcg	90 mcg	90 mcg

SIDE EFFECTS:

- Generally safe at dietary levels; no upper intake level has been established for K1 or K2.
- Excessive synthetic vitamin K (K3 or Menadione) can cause hemolytic anemia and jaundice, especially in infants, due to its pro-oxidant properties.
- Vitamin K antagonists (e.g., warfarin, dicoumarol) are medications that inhibit vitamin K function and are commonly used as anticoagulants, requiring careful dietary management of vitamin K intake.

Any Question